

Micro Project Report

Student Enrollment Form using JSON PowerDB

Submitted by: Shubham

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1. Introduction

This micro project involves creating a Student Enrollment Form that interacts with JSON PowerDB, a lightweight, NoSQL database service. The objective is to build a web form where student data can be inserted, updated, and managed efficiently using JPDB APIs.

2. Objective

The main objective of this project is to demonstrate the integration of front-end form handling with backend database operations using JSON PowerDB. The form includes validation, CRUD operations, and dynamic UI updates based on user input.

3. Technologies Used

- HTML, CSS, JavaScript
- JSON PowerDB
- JPDB API for database interaction
- FPDF for report generation (Python)

4. Project Overview

The form includes the following fields:

- Roll-No (Primary Key)
- Full-Name
- Class
- Birth-Date
- Address
- Enrollment-Date

Control Buttons:

- Save: To insert new student data if the Roll-No doesn't exist
- Update: To modify existing data if the Roll-No is already present
- Reset: To clear the form

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Functionality:

- On page load or Reset, only Roll-No is enabled.
- If Roll-No is new, Save and Reset are enabled.
- If Roll-No exists, data is displayed, Update and Reset are enabled.
- All fields are validated before data is saved or updated.

5. Conclusion

This project helped in understanding how to create dynamic web forms that communicate with a backend database using modern JavaScript and API techniques. JSON PowerDB offers a quick and effective way to handle data operations for lightweight applications.